

Low-Cost Bacterial Cellulose Aerogels for Lightweight Thermal Protection in Spacecraft Heat Shields

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Abstract

With the rapid increase in space exploration and the need for reusable spacecraft that survive extreme reentry, developing advanced, lightweight thermal protection systems (TPS) is critical. Current TPS architectures use dense, costly materials like reinforced carbon-carbon on the exterior, while the intermediate insulation layer, responsible for blocking heat from reaching internal structures, typically relies on ceramic fiber blankets/polymeric foams that degrade under thermal cycling, cause environmental concerns, and add mass. Aerogels present a promising alternative as silica aerogels offer superior insulation at low density, but their brittleness, poor mechanical strength, and vulnerability to thermal shock have limited their practicality. This project addresses the limitations of both existing TPS insulators and conventional aerogels by engineering a low-cost, bio-derived bacterial cellulose (BC) aerogel optimized for intermediate layer thermal protection. The material utilizes citric acid crosslinking combined with montmorillonite clay, polyvinyl alcohol, and other compatible additives to enhance functionality. Samples are fabricated through homogenization, hydrogel casting, thermal curing/esterification, ethanol solvent exchange, and lyophilization to preserve the structure. Performance is benchmarked against a commercial silica aerogel and ceramic fiber (TPS) through Differential Scanning Calorimetry (DSC), hybrid rheometry, and a Desktop SEM/microCT. DSC results reveal an average specific heat capacity of $\sim 2.83 \text{ J/g}\cdot\text{K}$, which is 3.4-fold higher than silica aerogel, alongside reduced shrinkage ($<10\%$) and thermal stability $>200^\circ\text{C}$ with no significant structural degradation. SEM and ImageJ analysis confirm a porosity of $\sim 94.2\%$ (pore size of $0.02\text{--}0.04 \mu\text{m}$), optimized for thermal insulation compared to ceramic fiber materials. Rheometry of crosslinked BC aerogels shows a $\sim 27\%$ increase in storage modulus relative to the unmodified baseline, indicating that the crosslinking step was effective. This work introduces a novel, scalable, and robust class of intermediate TPS insulators supporting sustainable space exploration.

Introduction

The reentry of spacecraft presents one of the most extreme thermal environments encountered in engineering practice. Convective and radiative heating rates produce surface temperatures of several thousand degrees Celsius and impose severe mechanical and thermal shocks on a vehicle's thermal protection system (TPS). To protect the internal structure and payload during ascent, hypersonic reentry, and on orbit operations, a layered TPS architecture must be created that balances high temperature surface protection (whether ablative materials or refractory ceramics) with low conductivity and low mass for intermediate insulation that prevents heat from reaching underlying structures during these transient heating events. As human missions to Mars and expanded space exploration become increasingly feasible, the demand for effective TPS has grown dramatically, highlighting their critical importance. A few examples clearly illustrate this trend: NASA's Artemis program aims to establish a sustained lunar presence by 2028, planned crewed missions to Mars in the 2030s are advancing rapidly, and the rapid expansion of commercial orbital platforms and space tourism ventures has significantly accelerated the timeline for next-generation TPS development. The design challenge is not only peak temperature

resistance but also repeatable/reusable performance under thermal cycling, mechanical loads, and the complex interaction that come with typical atmospheric reentry.

Currently, TPS solutions rely on two broad approaches: (1) high temperature, sacrificial or char-forming ablators with ceramic coatings for leading edges/stagnation points; and (2) intermediate and insulating layers which are commonly ceramic fiber blankets (e.g., alumina or silica fiber blankets) or polymeric foams that minimize heat conduction through fibrous microstructures and the stagnation of air, though their effectiveness degrades with thermal cycling and stress. Ablative TPS materials function by controlled decomposition: the most common example, phenolic-impregnated carbon ablator (PICA) or carbon-phenolic composites absorb heat through endothermic pyrolysis and char formation, with the char layer acting as an insulative barrier while mass loss carries away thermal energy. Leading edge materials such as reinforced carbon-carbon (RCC) rely on ultra-high temperature ceramics (UHTCs) like hafnium diboride or zirconium carbide that can withstand $>3000\text{ }^{\circ}\text{C}$ through radiative cooling, though these are dense ($5\text{--}6\text{ g}\cdot\text{cm}^{-3}$) and very prone to oxidation. Ablative systems are also essentially disposable, requiring refurbishment or replacement after each mission, which can lead to exorbitant costs. The Space Shuttle program, for example, spent approximately \$1-2 million per flight refurbishing thermal tiles, with turnaround times extending 2-3 months between missions. As the commercial space industry grows, projected to exceed \$50 billion annually by 2030, and reusable launch systems from SpaceX, Blue Origin, and Rocket Lab become standard, the economic burden of TPS maintenance scales proportionally, creating strong incentive for materials that reduce both mass penalties and refurbishment cycles. Ceramic fiber blankets and polymer foams, on the other hand, while lighter and reusable in some architectures, are often mechanically fragile, susceptible to progressive microcracking and oxidative/thermal degradation in space's harsh environment, and add considerable mass/logistical maintenance that burdens reusable vehicles. Therefore, engineering lighter, more robust intermediate insulators that are both compatible and retain insulating performance across cycling would reduce TPS mass, lower lifecycle maintenance, and enable more frequent reuse of spacecraft. Moreover, while thermal insulation and mechanical strength are primary considerations, an effective TPS must also withstand additional space-induced stresses such as cryogenic cycling, chemical degradation, and vibration or acoustic loads.

Aerogels, highly porous networks with $>90\%$ porosity, are widely recognized as the “gold standard” for low density insulation because of their unique pore architecture, which suppresses convective and gaseous conduction and dramatically lowers overall solid phase heat transport. Silica aerogels, in particular, are able to deliver thermal conductivities at or even below that of still air and can reach exceptionally low bulk densities $\sim 0.03\text{ to }0.2\text{ g}\cdot\text{cm}^{-3}$, making them attractive as TPS layers when mass minimization is critical. Three fundamental limitations have limited research in this area for aerospace usage: intrinsic brittleness/low fracture toughness, sensitivity to radiative heating and large thermal shock, and manufacturing costs that arise from the expensive solvent exchange and supercritical drying processes required to preserve their porous structure. Together, these constraints have hindered the widespread adoption of this nearly ideal material, despite its significant thermal and weight advantages. Addressing these limitations requires materials that combine the thermal performance of aerogels with improved mechanical robustness, thermal shock/radiation tolerance, and reduced manufacturing/environmental cost. These environmental costs range from energy-intensive sintering for production to drying that requires pressures exceeding 73 bar, consuming approximately 15-20 kWh per kilogram of material. Additionally, ablatives generate toxic phenolic and cyanide-containing decomposition products during reentry that

accumulate in the atmosphere and polymeric foams/blankets release volatile organic compounds and a heavy reliance on unsustainable resources.

A potentially promising, underexplored route is the development of bio-derived aerogels based on a bacterial cellulose precursor. Bacterial cellulose, which can come in many types and includes both grown and industrially produced versions, yields a high crystallinity nanofibrillar network that can be converted into an ultra-lightweight aerogel via controlled homogenization, solvent-exchange, and lyophilization. At first, this precursor may raise flammability concerns, but bacterial cellulose has many ideal characteristics, providing a sustainable, highly porous, and mechanically robust framework for advanced aerogel composites. The main way to create bacterial cellulose is by aerobic fermentation of the *Komagataeibacter* species, which polymerizes glucose in the growth medium into cellulose nanofibers. These fibers self-assemble into a highly pure, three-dimensional porous network free of lignin and hemicellulose. BC-derived aerogels have demonstrated favorable combinations of porosity, mechanical resilience, and thermal insulation in lab studies and their renewable origin offers sustainable advantages relative to petrochemical precursors. Additionally, their ability to be freeze-dried, which is much more cost efficient and easier to conduct compared to supercritical drying, allows the material to be much more scalable. Importantly, the BC network provides an intrinsic nanoscale skeleton that can act as a template for additional functionalization such as the incorporation of inorganic fillers or polymeric crosslinkers to tune and enhance mechanical/thermal behavior. To make BC aerogels practical for spacecraft TPS, three engineering challenges must be tackled simultaneously: (1) reduce the aerogel's solid phase thermal conductivity while maintaining integrity across harsh thermal cycles; (2) increase overall toughness and compressive resilience (including resistance to outside factors like cryogenic/radiative environments), to survive handling, panelization, and launch vibrations; and (3) establish fabrication approaches that lower overall cost and scale (exclusive reliance on methods like supercritical drying and complex sol-gel processing).

In this project, I pursue a material design strategy that combines citric acid esterification crosslinking, polyvinyl alcohol (PVA) reinforcement through an induced freeze-thaw cycle, and montmorillonite (MMT) clay incorporation to create a low cost bioderived BC aerogel tailored for intermediate TPS layers. Citric acid (a benign natural polycarboxylic acid) has been shown to form ester linkages with hydroxyl-rich polysaccharides and with hydroxyl-bearing polymers such as PVA, yielding strong covalent crosslinks that increase dimensional stability and compressive strength while avoiding harsher chemicals and being environmentally sustainable. PVA contributes to ductility and improves overall processability, especially after a freeze-thaw cycle (usually a 3-day cycle of freezing and thawing X times in a different environment to boost crystallization). Montmorillonite (a layered aluminosilicate) provides a plate-like reinforcement that forms tortuous conduction and radiative pathways, improving thermal stability and flame resistance when dispersed in a porous matrix. These additives, while rooted in different literatures and proven individually, have not been combined, and especially not experimented upon for aerospace TPS systems. This gap in existing research provides both a scientific opportunity and an engineering imperative to pave the path for this new class of materials and their respective applications. The following sections describe the material synthesis, characterization methods, experimental validation, and implications of this bio-derived composite for spacecraft thermal protection.

Methods

Before the actual experimentation, simulation was conducted using two softwares: COMSOL Multiphysics and SimScale. First, a $50 \times 50 \times 20$ mm aerogel tile (size varied by application) was produced through CAD in Fusion 360 and exported as a STL file. This was then imported to their respective locations. Thermal stack performance under reentry was evaluated using transient finite element thermal simulations. The objective of the simulation was to (1) quantify temperature handling of backface BC aerogel layers, (2) compare predicted thermal performance against baselines (commercial silica and ceramic fiber blanket), and (3) estimate required thickness/physical property targets for experimental design. The simulation utilized the transient heat conduction equation in three dimensions with additional coupled surface convection and radiation, $\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$, to ensure application accuracy. Surface radiation was represented using the Stefan Boltzmann relation ($q_{rad} = \epsilon \sigma (T_s^4 - T_{amb}^4)$) and linearized slightly for small temperature differences. Surface convection and thermal diffusivity were additionally applied to capture heat transfer at the material boundary and the rate of internal heat propagation for experimental validation. Where appropriate, temperature dependence/properties were implemented from literature data or from measured values, and if such data were unavailable, similar research was compiled and a rigorous estimation was made.

Simulations were first implemented in SimScale and used a three layer representative TPS geometry: (1) an outer high temperature surface (ablative or carbon carbon, thickness matching actual heat shield), (2) an intermediate insulating layer composed of the candidate material (BC aerogel or baselines), and (3) an internal structural backing used for both support and connection (aluminum or composite). Boundary conditions consisted of a time-varying heat flux applied to external faces and also radiative/convective loss at the low pressure, high temperature environment. The interior face was insulated except where the structural conduction in the payload was modeled. More advanced simulation was conducted using COMSOL, where mesh refinement was utilized across material interfaces to visualize more specific changes. A time step control was put in place to ensure adequate resolution of transient heating and other factors. Uncertainty and sensitivity analysis was conducted by varying key parameters (thermal conductivity, layer thickness, emissivity, applied heat flux, etc) across plausible ranges to quantify influence on back face temperature mass. Simulation consisted of tests run for thermal, mechanical, and basic properties like density and porosity, but also a few environment-specific factors were slightly tested including radiative resistance and reaction to cryogenic cycling through temperature manipulation. These outputs directly informed experimental design by identifying target ranges for thermal conductivity, optimal thickness, and other traits, building the baseline for models. Additionally, molecular dynamics simulations with LAMMPS software were carried out by writing a custom input script defining a bacterial cellulose network, interatomic potentials, and crosslinking interactions, followed by applying controlled thermal loading conditions. The resulting simulations were then used to analyze nanoscale bond behavior and stress distribution, providing insight into how crosslinking and material enhancers improve mechanical stability at the microscale.

The experimental program combined materials synthesis, accessible low equipment processing, professional lyophilization, and lab grade characterization. Four compositional series were fabricated (P1-P4, described below) and screened by a suite of characterization methods (Differential Scanning

Calorimetry (DSC), TA Instruments Discovery HR-2 Hybrid Rheometer, and Scanning Electron Microscope (SEM)).

Materials

Bacterial cellulose feedstock was obtained as commercial nata de coco (Kapuso, Asian Food Trading Co.). Polyvinyl alcohol (4%), citric acid (ACS reagent grade), agar powder, and montmorillonite clay (MMT, sodium form), were purchased online from Amazon. Ethanol (>95%), NaOH, and distilled water (DI) were used for cleaning, solvent exchange, and overall pH adjustments.

BC extraction/Slurry preparation

Nata de coco cubes were first rinsed in distilled water thoroughly (around 340 grams per bottle, creating 8 samples), until immersed in a NaOH bath for an hour to remove residual sugars, then neutralized by repeated washes with DI water until the pH reached ~7, and kept over night. The cubes were then homogenized in a blender with additional DI to produce a homogenous BC slurry. The slurry was then stored at about 4 °C for approximately 48 hours.

Formulation and gel casting

Four batches were planned before creation: P1 (baseline): BC + agar, P2: BC + agar + PVA + citric acid (crosslinker), P3: P2 formulation + low MMT loading, P4: P2 formulation + high MMT loading. An agar solution was prepared by dissolving agar powder in DI water (2 grams of agar for 100 ml of water) and stirred completely before ramping heat. Then depending on batch, for P1 and P2, the solution was heated to 90 °C in a beaker submerged in a pot of water on a stovetop for approximately 5 minutes until a clear solution appeared. For P3 and P4, the solution was created by adding the same agar amount and distilled water, but then (1 gram of clay for P3 and 2.5 grams for P4) and stirring completely. The heating process was repeated. In the meantime, 75 mL of 4% PVA was placed in a beaker, and 3 grams of pure citric acid was stored. To prevent quick gelling after heating, right after the solution was released from the heat, we mixed in the PVA, stirred for a minute, then citric acid powder was added and stirred to dissolution. Composite precursor suspensions were then placed in equally sized 50 × 50 × 20 mm sample holders with slight spacing to account for shrinkage and then placed in the refrigerator for ~45 minutes to gel.

Freeze thaw strengthening and esterification

Cast hydrogels then underwent a freeze thaw protocol to promote PVA crystallization and network consolidation: samples were frozen at -20 °C in a freezer and thawed at ambient temperature in cycles over approximately three days (12 hour cycles for each). For esterification, 4 samples were chosen (2 P2, 1 P3, 1 P4) and for these subsets, they were partially dried (slightly damp state) for a day. One of those samples was then chosen for extra citric acid immersion, soaking in a 15% bath (15 grams for 100 mL) for 20 minutes before being dried. They were then placed in an oven at 80-90 °C for ~2 hours to promote citric acid esterification, and then the samples were rehydrated by DI water exchanges every few hours for 24 hours to remove unreacted residues and solvent byproducts.

Drying protocols and lyophilization

Two lower equipment drying techniques were developed for initial screening and replication, and then professional lyophilization was utilized for final samples: (1) Dry ice assisted sublimation protocol: Samples were placed in shallow trays and rapidly frozen by immersion in a dry ice/ethanol bath until core temperature was below 40 °C. Frozen samples were then transferred to a makeshift rudimentary vacuum chamber connected to a mechanical pump to sublime the solvent. The samples were warmed slowly in several stages to room temperature to avoid any recondensation. (2) Ambient pressure solvent exchange drying: After graded solvent exchanges (30%, 60%, 75%, 95% ethanol, each step for a certain time), samples were dried in a controlled-humidity enclosure and allowed to rest, with slight extra heating mimicking thermal annealing cycles to reduce any capillary stress/shrinking. (3) The main drying process that created the final samples was conducted at Rice University, where selected samples were frozen in liquid nitrogen (around -200 °C), transferred to a Milrock benchtop freeze dryer and then sublimated (solid to gas) under standard protocol for around 3 days. The lyophilization preserved pore architecture and served as reference for final characterization. Batch production counts: P1 (3 samples, 2 freeze dried, one not), P2 (3 samples, two esterified, one with initial dispersed citric acid powder and one subjected to an additional 10% citric acid soak for 15 minutes prior to esterification), P3 (3 samples: one esterified, two non esterified), and P4 (3 samples: one esterified, and in total, in the professional freeze dryer 8 samples were prepared for the first batch. Each fabricated specimen was assigned a unique sample ID and documented in the project log (processing route, dates, operator).

Characterization Methods

At least three independent measurements were performed/recorded on each sample with available instruments that matched or approximated the desired measurements for the project's goals. DSC characterization was performed on a TA Instruments DSC at the Rice Biomaterials Lab using hermetically sealed Tzero aluminum pans. Sample masses were weighed to around 3-5 mg. Then they were loaded into the DSC cell and heated by the protocol: -40 °C to 250 °C at 20 °C·min⁻¹. The sapphire reference method was used to calculate specific heat capacity (C_p). Raw heat flow (μ W) was mass-normalized to obtain the q_{norm} in W·g⁻¹ and converted to C_p ($C_p = q_{norm}/\beta$). The cumulative heat (the total heat exposure) was evaluated by integrating the baseline heat flow curve formed by DSC data points. Additional cycles were run to quantify any sample to sample variability and validate findings. For SEM and general property analysis, lyophilized samples were mounted on aluminum stubs and sputter-coated with a 5–8 nm Au/Pd layer to minimize charging. Imaging was performed at 5-15 kV accelerating voltage with sizes ranging from 5-10 nm. Multiple magnifications were recorded to measure porosity, crosslinking, structure, and assess homogeneity. Porosity quantification used ImageJ with the pipeline: convert to 8 bit grayscale, apply Otsu automated thresholding, remove outliers and morphological errors, measure area fraction (pore area ÷ total area), calculate porosity as a percentage. From this, bulk density was both calculated through porosity relationship and measured with an analytical balance (± 0.01 mg), as sample volume was measured (thickness × area) and then reported in g·cm⁻³. Shrinkage was calculated by comparing pre and post drying dimensions. Finally, mechanical testing samples were prepared as thin aerogel disks by carefully sectioning lyophilized monoliths using a razor blade to achieve uniform thickness and parallel faces, followed by diameter trimming to ensure alignment with the hybrid rheometer's plate geometry. Samples were stored in a desiccated environment prior to testing to minimize moisture uptake and

manage structure. Quasi-static axial compression testing was performed on a TA Instruments Discovery HR-2 hybrid rheometer at Rice which allowed for detailed results. Three lyophilized aerogel disks were tested: one crosslinked sample (P3 formulation) and two baseline samples. Each disk was placed between 8.0 mm stainless steel parallel plates with Peltier temperature control maintaining 25.00 ± 0.02 °C throughout the test. The crosslinked sample had an initial thickness of 6.49 mm, and both baselines measured 6.74 mm. All three samples were compressed to approximately 50% of their original thickness, taking roughly over five minutes per test. The initial compressive modulus (E_0) was determined by linear regression of the stress–strain curve over the 0–5% strain interval, following the relation $\sigma = E_0 \cdot \epsilon$, where σ is the compressive stress and ϵ is the engineering strain. The tangent modulus ($d\sigma/d\epsilon$) was computed as the numerical derivative of stress with respect to strain at each data point to track instantaneous stiffness evolution throughout compression. Total energy absorption (mechanical toughness) was calculated by integrating the stress–strain curve using the trapezoidal rule. Specific energy absorption was obtained by normalizing volumetric energy density through bulk density. Compression regime transitions were identified from inflection points in the tangent modulus profile, consistent with the Gibson-Ashby cellular solids model for open-cell foams. Additionally, oscillatory loading was applied to evaluate viscoelastic response and storage modulus under low-strain conditions, mimicking TPS service environments.

Results and Discussion

Note: numbers in square brackets refer to figure numbers in the Visual Model.

The thermal characterization of the BC aerogels shows a distinct performance state relative to the two benchmarks: commercial silica aerogels and conventional ceramic fiber TPS insulation. This suggests that the BC compositions occupy a different region of TPS material space, one that favors transient energy storage and resistance to thermal cycling over maximizing lower steady state thermal conductivity [33–42]. Measured specific heat capacities for the BC series ranged from $2.71 \text{ J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$ for the baseline (P1) to $3.12 \text{ J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$ (P3) [33], with an average of $\approx 2.9 \text{ J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$. Such C_p values are high on the spectrum for ultralight aerogels and directly increase the material’s volumetric thermal inertia; this elevated heat capacity makes it very beneficial for delaying heat transmission during short, high-flux events like reentry. By contrast, representative silica aerogel C_p values from both a store-bought model and literature measured roughly $\sim 0.84 \text{ J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$ [33]. Typical ceramic fiber products for heat shields have C_p values spanning about $0.8\text{--}1.1 \text{ J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$, depending on the specific formulation. The result showcases that BC aerogels store $\sim 3\text{--}4\times$ more thermal energy than either silica aerogel or standard ceramic fiber blankets, a difference that translates directly into transient thermal buffering capacity in high heat flux events such as atmospheric reentry.

For thermal conductivity, a different order is observed, but with key tradeoffs. The silica aerogel composite reports a conductivity range of $\sim 0.013\text{--}0.025 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ at room temperature [34], while many blanket versions are around $0.024 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, illustrating silica aerogels’ reputation as one of the lowest-conductivity solids. In contrast, the ceramic fiber exhibits a much higher conductivity ($0.06\text{--}0.2 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) [34], which varies for industry products and standard densities. Based on calculations utilizing measured porosity and literature values for the conductivity of air and solid cellulose, the conductivity of enhanced BC aerogel is estimated at $k \approx 0.035 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ [34], which is higher than the silica but substantially lower than most ceramic fiber options. This result is most likely caused by the addition of enhancers, as the BC material operates in an intermediate band, where it sacrifices some

steady-state insulation, but remains almost as light and much stronger. I attribute this to three microstructural effects: (1) preserved nanoscale porosity that keeps density low, (2) PVA crystallites formed during freeze–thaw that increase toughness and dissipate mechanical energy, and (3) low loading of MMT clay, which increases path length for conductivity without collapsing pore volume. Literature on PVA freeze–thaw gels and citric-acid crosslinking supports these mechanisms and the balanced results. It specifically forgoes an extremely low insulating extremity for markedly higher C_p and lower mass per unit area than most ceramic options. A simple conduction-only comparison holding area and ΔT constant indicates a BC layer would need to be $\sim 1.4\text{--}1.5\times$ thicker than a silica blanket to match steady-state thermal resistance ($R = L/k$). But this omits time-dependent physics, which is important under reentry conditions: the BC materials' high C_p and low thermal diffusivity cause them to absorb and delay energy transmission in ways a low- C_p silica layer cannot, often producing lower peak back-face temperatures for BC models in transient simulations [13,46], highlighting a key efficiency gap. To quantify the time-dependent response, thermal diffusivity (α) was calculated from measured/derived properties using the relation $\alpha = k/\rho C_p$ (k is the effective thermal conductivity, ρ is the bulk density, and C_p is the measured specific heat capacity). Using the measured densities calculated from both porosity and volume calculations (P1: $\sim 0.0674\text{ g}\cdot\text{cm}^{-3}$; P2: $\sim 0.0697\text{ g}\cdot\text{cm}^{-3}$; P3: $\sim 0.0721\text{ g}\cdot\text{cm}^{-3}$) and measured C_p values, I obtained estimated thermal diffusivities ranging from $\alpha \approx 1.9 \times 10^{-7}\text{ m}^2\cdot\text{s}^{-1}$ to $1.56 \times 10^{-7}\text{ m}^2\cdot\text{s}^{-1}$. These values are exceptionally low relative to metals ($\sim 10^{-5}\text{ m}^2\cdot\text{s}^{-1}$) and indicate very slow bulk temperature penetration, a desirable attribute for intermediate TPS layers that must buffer reentry heating pulses. However, direct experimental determination of α using a laser flash method is planned in future work to capture potential anisotropy and interfacial effects, which were not fully resolved by property-based estimation.

Differential scanning calorimetry (DSC) traces, measured at a heating ramp of $20\text{ }^\circ\text{C}\cdot\text{min}^{-1}$ ($\beta = 0.333\text{ K}\cdot\text{s}^{-1}$) between -40 and $250\text{ }^\circ\text{C}$, clearly highlight the differences in formulations [35,36]. All samples maintained a near-flat baseline heat flow of $\sim 0.6\text{--}1.0\text{ W}\cdot\text{g}^{-1}$ from room temperature through $200\text{ }^\circ\text{C}$, and withstood energies of $\sim 300\text{--}320\text{ J}\cdot\text{g}^{-1}$ across a full heating cycle. However, the similarities appear to end there. The baseline P1 (BC + agar) provides the lightest, most open network, with a calculated porosity of 94.4% from ImageJ analysis and a density $\approx 0.0674\text{ g}\cdot\text{cm}^{-3}$. Though the shrinkage was slightly higher in this set, it was still relatively low ($<15\%$) and it performed respectably in thermal buffering ($C_p = 2.71\text{ J}\cdot\text{g}^{-1}\cdot\text{K}^{-1}$). The modest shrinkage suggests the freeze-drying protocol preserved the fibrillar network well enough for actual application, as minimizing shrinkage is critical to maintain designed thicknesses and predictable R values. In addition, two lower-cost drying routes were also evaluated prior to lyophilization to assess accessibility. The dry ice–assisted sublimation protocol, in which samples were rapidly frozen in a dry ice bath and sublimated under a rudimentary mechanical vacuum with ambient drying, produced aerogels with visibly higher shrinkage and reduced uniformity compared to professionally lyophilized samples. However, the basic porous architecture was retained, yielding a much lower-quality aerogel at roughly $600\times$ less cost, highlighting the opportunity to move away from this otherwise inaccessible supercritical drying process. The ambient pressure solvent exchange method, which involved graded ethanol exchanges followed by controlled evaporation, resulted in the greatest shrinkage and some visible densification at sample edges, yet still produced lightweight monoliths with measurable insulating properties, demonstrating a potential route to reduce equipment needs. While both alternative methods clearly fell short of the lyophilized reference, their viability at multiple levels underscores a key practical advantage of the BC platform and the unique fabrication processes it enables. But the key problem comes

with strength, as P1's mechanical and endurance response is clearly the weakest of the set, which is shown in the thermal cycling tests [37], as it was first to begin degrading and lasted the fewest cycles. Additionally, the unreinforced fibrillar network is somewhat brittle under compression, a key problem. Compression testing on the hybrid rheometer [37] confirmed this quantitatively, as baseline BC aerogel samples (P1 formulation) exhibited an initial compressive modulus of only 16.3–22.7 kPa over the 0–5% strain range, with peak compressive stresses of just 9.4–13.3 kPa at approximately 50% strain and total energy absorption of 2.0–3.2 kJ/m³ [39]. The stress-strain curves that were derived from baseline formulations (P1 variations) almost showed a linear rise without any clear regime transitions, indicating that the weakly hydrogen-bonded fibrillar network lacks the structural integrity needed to resist pore collapse [40]. This severely limits its ability to tolerate handling, panelization, and thermal cycling without early microcracking.

Adding polymeric reinforcement and the citric acid crosslinker (P2: BC + agar + PVA + citric acid) shifts the balance toward toughness while maintaining low mass. P2 had an increased C_p of 2.90 J·g⁻¹·K⁻¹, though with a slightly increased bulk density (≈ 0.0697 g·cm⁻³). P2 had a measurable improvement in stiffness vs. P1, consistent with PVA crystallization during freeze-thaw and initial ester link formation. The DSC trace for P2 shows a beginning of the broad endothermic feature observed in the P3 formulation, indicating that even moderate citric-acid content can contribute to buffering and network relaxation phenomena [36]. The P3 formulation, which uses the same base chemistry as P2 but includes a low amount of clay, and an additional sample made with an additional citric acid soak prior to esterification, is the clear standout shown in Figure 36. P3 recorded the highest C_p (3.12 J·g⁻¹·K⁻¹) [33] and introduced a broad endothermic region (130–180 °C) where peak temperature was clearly reduced (each gram absorbed around 90–120 J·g⁻¹ before the temperature rose normally), and had the best combined mechanical/thermal behavior: density remained ultra-low at ≈ 0.0721 g·cm⁻³, shrinkage remained below 10%, and hybrid-rheometer testing showed a dramatic improvement in compressive mechanical performance. Quasi static compression at 25 °C using the 8.0 mm parallel plate geometry revealed that the crosslinked BC aerogel achieved an initial compressive modulus (E_0) of ~ 179.2 kPa over many trials in the 0–5% strain interval representing a 7.9-fold increase over the stronger baseline (22.7 kPa) and an 11.0-fold increase over the weaker baseline (16.3 kPa) [39]. The modulus further increased to 194.6 kPa in the 5–10% strain range, indicating strain-stiffening behavior as crosslink junctions introduced additional load bearing pathways under deformation, in direct contrast to the previous baselines, which exhibited flat and even softening modulus in the same range. Peak compressive stress for the crosslinked formulation reached an average of around 59.6 kPa at approximately 33–35% strain, with some tests showing slightly over 65 kPa, with a corresponding peak axial force of 3.0 N, compared to just 9.4–13.3 kPa and 0.47–0.67 N for P1 and P2 [39,40]. Upon unloading from 40% compressive strain, P3 exhibited approximately 91% $\pm$ 4% height recovery, indicating strong elastic resilience with minor damage. Most notably, the crosslinked sample's stress-strain curve displayed the three regimes predicted by the well-known Gibson-Ashby cellular solids model [40]: (1) a linear elastic region from 0 to $\sim 8\%$ strain where cellulose nanofibrils bend reversibly, (2) a plateau and progressive pore collapse regime from ~ 8 to $\sim 35\%$ strain where cell walls sequentially buckle with broad stress maximum, and (3) a densification onset above $\sim 35\%$ strain where collapsed solid material begins compressing against itself. The emergence of these regimes confirms that the lyophilization protocol preserved a true aerogel cellular architecture in the crosslinked samples and that the material functions as an engineered porous solid. By utilizing a progressive pore collapse in Region II rather than the automatic brittle fracture typical of silica

aerogels at just 5–15% strain, P3 helps represent a failure mode that is highly desirable for TPS intermediate layers, preventing sudden structure loss. Tangent modulus analysis ($d\sigma/d\epsilon$) further confirmed this organized behavior, with the crosslinked sample exhibiting a peak instantaneous stiffness of ~ 243 kPa in the elastic region that remained elevated above both baselines throughout the entire compression range, even during the pore collapse softening zone [42].

The broad endothermic feature is consistent with reversible energy-dissipation mechanisms reported in polymer–nanofibril and bio-derived aerogel systems, where ester linkages and hydrogen bonding gradually relax rather than undergo a sharp phase transition. This behavior suggests a distribution of activation energies within the network, with the observed endotherm interpreted as an emergent property of the combined crosslink structure. The low clay amount allowed for the “rerouting” of intense heat conduction, allowing for more effective thermal insulation. Most importantly, P3 endured multiple thermal cycles with little fracture, a key ability needed in the constant flux of space environments, and was very reproducible with a cycle-to-cycle C_p variation $< 5\%$ and post-cycle microscopy showing little to no warping. These observations indicate the extra citric acid soak substantially increased effective ester crosslink density and/or improved crosslink distribution, producing a flexible yet cohesive network that both (a) creates additional energy-absorbing processes (the broad DSC endotherm) and (b) allows fibrils to slightly deform rather than fracture under stress. The rheometer data provides confirmation of this mechanism: the crosslinking enhancement ratio peaks at $9\text{--}11\times$ relative to the baseline average in 20–30% range, the place where most pore collapse takes place and where crosslinking is critical for resistance. Total energy absorption for the crosslinked aerogel (P3) reached 20.4 kJ/m^3 [41] with baselines having a specific energy absorption varying around 0.293 kJ/kg at only 0.07 g/cm^3 bulk density. The energy absorption efficiency reached much higher levels for the crosslinked aerogel, but this increase is not fully needed. P3 highlights a tradeoff in function, because rather than mimicking an ablative that can absorb massive amounts of energy, it relies on a very strong thermal resistance to prevent heat from even reaching the structure.

By contrast, P4 (the high-clay loading formulation) exposed the limits of overcrowding the fibrillar matrix with platelets. The original sample before lyophilization was observed to be the softest, likely due to the clay's high water retention, and while montmorillonite platelets can increase tortuosity for heat paths and provide crack-bridging (extrinsic toughening), excessive clay loading can reduce porosity beyond the optimal window and produce a strikingly more brittle structure. This demonstrates the percolation-like behavior of nanoplatelet fillers, as beyond a threshold, filler aggregation reverses any gains. The tangent modulus showcases that the crosslinked formulation exhibits a U-shaped stiffness across the range (high initial stiffness, organized softening to gradually reduce stress, and re-stiffening during densification), while the baselines show no such structural transition as fibrils slide against each other [42]. Thermally, P4 did not show the same magnitude of the endothermic buffering feature as P3 and did not reach the same C_p (~ 2.9 range). It seems the clay appears to trade some energy-absorbing capacity for conduction-path disruption.

To place these results in the context of existing baselines and factors needed for reentry conditions, monolithic silica aerogels have densities of around $0.04\text{--}0.20 \text{ g}\cdot\text{cm}^{-3}$, infrared emissivity ranging from 0.9 to 0.92, a low coefficient of expansion $\sim 2.5 \times 10^{-6}$, solid vacuum and outgassing behavior, and shrinkage of 15–25% during processing. Their brittleness, low mechanical strength, and poor conformal adaptability to specific geometries limits their widespread applicability. Similarly, TPS ceramic fibers have typical

bulk densities in the range of ~ 0.09 to $0.2 \text{ g}\cdot\text{cm}^{-3}$, with a lower infrared emissivity ~ 0.6 , a coefficient of expansion of $\sim 4 \cdot 10^{-6}$, and lower linear shrinkage (3-5%). However, their manufacturing process is highly unsustainable and environmentally damaging, releasing gases suspected to be human carcinogens. Additionally, their inherently low tensile and compressive strengths make them susceptible to mechanical damage, while higher-temperature devitrification triggers crystallization that can lead to the potential failure of the full TPS assembly. Compared with these, the BC aerogels, especially the P3 formulation, create a material that is both lighter than typical ceramic fibers and provides better thermal buffering with less shrinkage ($<10\%$) than silica aerogels. While commercial silica aerogel composites have a modulus of around 100-200 kPa and a slightly lower density, their automatic collapse under pressure, rather than gradual change shown by P3, and brittle structure make it too delicate for application. Compared to ceramic fiber blankets, P3's kPa is on the higher end of the range (50-175) at 10% strain, the perfect balance between the two alternatives [39,40]. Additionally, while ceramic fiber blankets may be more flexible and thus mechanically adequate at higher densities, they suffer from fiber pullout degradation and devitrification that progressively weakens the structure, a pathway that P3's crosslinked network inherently resists due to the distributed nature of its ester bond and PVA crystallite junctions.

Notably, the addition of a small amount of montmorillonite clay to bacterial cellulose reduces infrared emissivity. This is beneficial for intermediate layers, as the goal is to minimize heat transferred through layers. MMT-augmented BC is thus much more effective than silica aerogel for infrared emissivity. If confirmed across larger panels and temperature ranges, reduced emissivity combined with high C_p could further lower integrated heat transfer during reentry pulses. Through simulation, the calculated coefficient of thermal expansion (CTE) of $\approx 1.2 \cdot 10^{-6} \text{ K}^{-1}$ P3 is not only lower than the two baselines, but is also similar to the CTE of ceramics, showcasing clear compatibility between TPS layers (similar CTE reduces thermal stress at interfaces). Finally, the citric acid network and the MMT nanoplatelets create a physical barrier that significantly reduces gas permeability, resulting in limited outgassing, while the PVA acts as a binder to withstand pressures faced by vacuum conditions without delaminating. Both of these properties are perfect for intermediate thermal layers. Early indications from limited outgassing checks are promising, but ASTM E595-style testing is required to validate long-duration vacuum compatibility. All P2, P3, and most P4 samples withstood heating ramps of $20 \text{ }^\circ\text{C}\cdot\text{min}^{-1}$ up to $250 \text{ }^\circ\text{C}$ without observable deformation, directly matching the operating window of intermediate spacecraft TPS layers, which are typically exposed to temperatures between $\sim 150\text{--}250 \text{ }^\circ\text{C}$ in reusable, shuttle-type vehicles. While the samples do not beat the baselines in every category, they hold key tradeoffs that blend the benefits from both sides, creating a composite that can uniquely face the conditions of space.

COMSOL simulations evaluated properties that were inaccessible for lab characterization, including radiative heating and cryogenic cycling. The open crosslinked, nanofibrillar network of BC aerogels disperses radiative flux efficiently, while the low solid fraction mitigates thermal strain that typically would compromise a silica aerogel under cycling. With the addition of an IR opacifier like carbon black, simulation predicts even greater radiative attenuation, allowing the BC composite to outperform most baselines and provide clear UV and thermal radiation protection. The model also predicted stability under rapidly changing temperatures (testing around $-100 \text{ }^\circ\text{C}$ to $20 \text{ }^\circ\text{C}$), highlighting capability in unique environments. Microstructural characterization (SEM with ImageJ quantification) revealed that all lyophilized BC samples had a porous structure with interconnected networks throughout the material [slide 13]. The measured porosity from Otsu analysis (94.4% for P1 and 92.7% for P3) show a diameter range of $\sim 20\text{--}40 \text{ nm}$. The narrow pore-size range suggests relatively uniform freeze-drying that limited

the formation of large voids that would otherwise concentrate stress in localized regions. Additionally, the connectivity introduced by the citric acid network helps distribute strain and reduce crack initiation, which is particularly beneficial in TPS stacks [40]. The mechanical results corroborate this microstructural interpretation, as the pores and network observed in the SEM matches the differences in characterization. Had the freeze-drying produced large voids/non-uniform pore distributions, the compression curve would have exhibited irregular stress drops and localized failure events. The combination of PVA crystallization from freeze-thaw cycling and esterification that builds the crosslinked structure provides the perfect backup for extreme stress and complements each other's strengths.

Taken together, the measurements indicate a shift in design objective. This perspective distinguishes the present work from much of the existing bio-aerogel and bacterial cellulose literature, which has largely emphasized low conductivity or mechanical reinforcement in isolation. This work integrates the benefits of both, prioritizing reliability and applicability over optimization at the extremes. Limitations do exist, however, and they are primarily specific to spacecraft-relevant environments: (1) the materials have not yet been evaluated under plasma or arc-jet conditions where surface chemistry, catalytic heating, and ablation may alter response; (2) exposure to atomic oxygen, UV radiation, and charged particle environments typical of low orbit has not been assessed; (3) and resistance to micrometeoroids, debris, vibration, and long term aging remain to be determined. Despite these constraints, the results carry important implications for both materials and aerospace research by demonstrating that sustainable, bio-derived aerogels can be deliberately engineered to access a performance area traditionally dominated by inorganic systems, and by highlighting how transient thermal response tunability is an underexplored region. More broadly, the accessible processing and design framework developed here suggests future applicability in areas including window and façade insulation that buffers solar transients, cryogenic insulation for propellant storage, thermal management layers for batteries and power electronics, and more, which would all drastically reduce energy consumption in use.

Conclusions

This project aimed to develop a low-cost, bio-derived aerogel architecture tailored for the intermediate layer of reusable spacecraft thermal protection systems. By combining a renewable cellulose precursor with benign chemical crosslinking and modest nanoplatelet reinforcement, I produced a set of composite materials whose performance priorities, enhanced energy storage, lightweight nature, controlled heat propagation, and cyclable mechanical resilience are explicitly aligned with the demands of repeated reentry and low-maintenance spacecraft architectures. By integrating both experimental and computational work, this research establishes a material design pathway in which processing choices tune a material's time-dependent thermal response and mechanical endurance without relying on expensive, unsustainable, or environmentally harmful manufacturing. Beyond demonstrating a promising materials concept, the work advances two broader research directions. First, it establishes that sustainable precursors and low-impact chemistries are capable of achieving application-relevant performance once considered exclusive to conventional inorganic materials, and that accessible processing routes like dry ice sublimation can provide practical alternatives to steps such as supercritical drying, a major barrier in material fabrication. Second, it introduces transient thermal response (volumetric energy storage and relaxation behavior) as a primary design axis for intermediate TPS materials, optimizing and providing a new outlook for tradeoffs in reusable systems.

Looking ahead, four key directions emerge for future development. (1) High temperature stabilization: carbonization and the incorporation of nanoparticles (for example, zinc borate or ammonium phosphate for flame resistance, and nanoparticulate opacifiers or carbides for improved radiative control and ablation resistance) can increase operating temperature, oxidation resistance, and emissivity, which are relevant to space environments. (2) System-level TPS integration/testing: instrumented panel tests that replicate an RCC (or UHTC) outer surface + BC aerogel intermediate layer + structural backing (with representative adhesives) are required to measure compatibility and back face temperature under arc-jet and plasma heat flux. (3) Conformal geometries: develop processing routes for curved and conformal aerogel components (molded monoliths, cast-in-place gels, or conformal sheets) and assess thermo-mechanical performance. (4) Higher-fidelity multiphysics modeling: extend simulations to couple radiative transfer, anisotropic diffusion, and other properties to build models and calibrate for optimal performance. Taken together, these next steps will validate environment-specific applications of this model, while also broadening the material's applicability beyond spacecraft.

The approach demonstrated here points to a novel, scalable class of insulating materials that can either complement or replace incumbent intermediate TPS solutions while also pointing to potential cryogenic systems, more reliable and safer heat management, and more. This work therefore establishes both a practical materials solution and a sustainable design paradigm for both high-performance thermal management and future innovative space technology.

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